

Oh My Gas! A Time Series Analysis of California Gas Prices

Federica Domecq Lacroze

MATH155 - Time Series

Professor Gabriel Chandler

Pomona College

1 Introduction

It comes as no surprise, that given its size, population and distribution, with 27 million drivers, California has more licensed drivers than any other state. In fact, 88% of the population that is of driving age is licensed. Coincidentally, at this point in time, California is also the state with the highest gas prices. It follows that the state of California's gas prices is of peak concern. Car owners and frequent long distance commuters wonder each day how much they will be paying for their next gallon of gas.

The analysis of California's gas prices is particularly intriguing due to the state's unique economic and environmental landscape, which significantly impacts fuel costs. Literature on this topic highlights various factors contributing to the state's higher-than-average prices. These include higher state gas taxes, clean air policies such as the cap-and-trade program for greenhouse-gas emissions, and the mandate for cleaner-burning gasoline (Borenstein, 2023). Additionally, California's nearly self-sufficient market with limited external supply and the impact of global events like Russia's invasion of Ukraine on crude oil prices make the state's gasoline market particularly volatile (Hailemariam, 2019) and interesting for economic analysis.

In the remainder of this paper we will attempt to apply a SARIMA model, that is a Seasonal Autoregressive Integrated Moving Average model, to California gas-price data in order to understand the fluctuating patterns of this highly-demanded commodity, aiming to predict future trends. In section 2, we will proceed by detailing our data set, and how it was attained. In section 3, we will present SARIMA model's methodology, including model selection and diagnostic check on our gas-price data. In our final section, 4, we will present the implications of our results and findings. We aim to provide a comprehensive overview of California's gas price trends.

2 Methods

For this time series analysis, we scraped the EIA's (U.S. Energy Information Administration) data base for California's weekly gas prices. As indicated by their documentation,

every Monday at 8:00 AM local time, EIA collects information on retail prices for regular, midgrade and premium grades of gasoline from a sample of retail gasoline outlets accross the United States. They share a weekly survey designed to retrieve data on the price offered at the pump including state taxation to consumers for the different gasoline grades offered for the self-serve price in areas having only full-serve. Prices are collected via different mediums and automated error checks are run during the time of data collection and processing. Those of dubious nature are checked with the original respondents. In the case of missing data, imputation (the process of replacing missing data with estimated values based on other available information) is carried out to estimate prices for outlets that couldn't be reached. The weekly price data from these samples are used to calculate volume-weighted average gas price estimates at national, regional, city and state levels for all the different gas grades and formulations.

California needs to produce reformulated gasoline to meet the specifications defined by the California Air Resources Board. Amongst the different reformulated gas options, we made the decision to focus on the most commonly sold type of gas for vehicles: unleaded gasoline, regular gas, marked with an octane rating of 87.

Our sample is comprised of the price in USD to 3 decimal places of California regular reformulated gas retail prices sampled weekly starting January 2nd 1995 up until March 18th 2024, that is 1525 observations. We will proceed to perform our SARIMA Data Analysis using R.

3 Data Analysis

$SARIMA(p, d, q) \times (P, D, Q)_S$ time series are forecasting models that take into account past values (autoregressive, moving average) and seasonality patterns to predict future values. These models can most generally be summarised using the backshift ($B(x_t) = x_{t-1}$) and differencing ($\nabla(x_t) = x_t - x_{t-1}$) operators in the equation $\Phi(B^S)\phi(B)\nabla_S^D\nabla^d x_t = \Theta(B^S)\theta(B)\epsilon_t$. In this model, with season S the Φ indicates the coefficients corresponding to the seasonal AR model, ϕ indicates the coefficients corresponding to our regular AR

model, Θ corresponds to our seasonal MA structure and θ to our regular MA structure.

Before using this sort of model to analyse and forecast our gas price data, we must see if our data meets the basic assumptions needed to perform such kinds of analyses – weak stationarity. The assumption of stationarity is met when $\mu_t = \mu$ and $\gamma(t, t + h) = \gamma(h)$, that is the mean of our process and the covariance structure is independent of time. The covariance we calculate therefore is dependent on the distance/lag between observations instead. Stationarity implies that the statistical properties of the process of generating time series data (mean, variance, autocorrelation) are constant over time. This stability is essential for the SARIMA model to make reliable and accurate predictions, as it relies on the assumption that patterns observed in the past will continue to be observed in the future.

Let us plot our raw time series data and determine whether it seems to adhere to heteroscedacity, ($\gamma(t, t + h) = \gamma(h)$), and whether the data seems to remain close to a constant ($\mu = \mu_t$) or deviates over time.

In Figure 1, we can see that as time goes on, variability seems to increase, not meeting one of our assumptions, therefore maybe if we take the natural log of our data, a common statistical transformation, we can deal with this issue and observe data that meets our heteroscedacity requirements, see Figure 2.

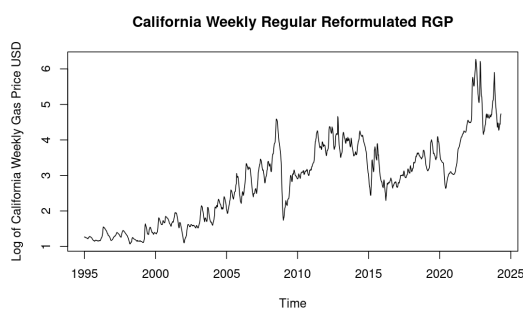


Figure 1: Our original time series data without any adjustment

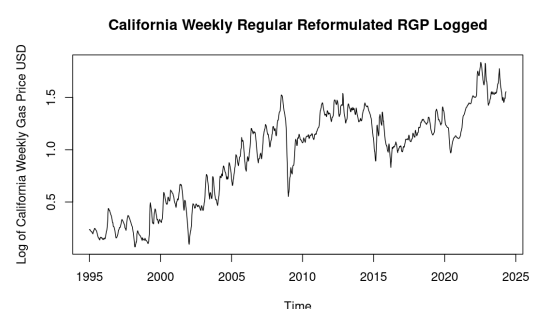


Figure 2: Taking the log of our time series data

Potentially not perfect, but definitely more consistent! We can see that there is a very evident, somewhat linear, upwards trend over time so potentially differencing our data once will lead to stationarity as defined by mean, Figure 3.

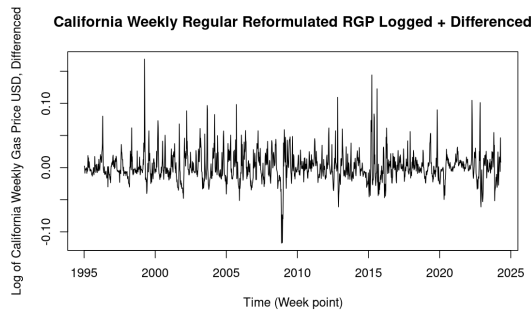


Figure 3: California Retail Gas Prices logged and differenced

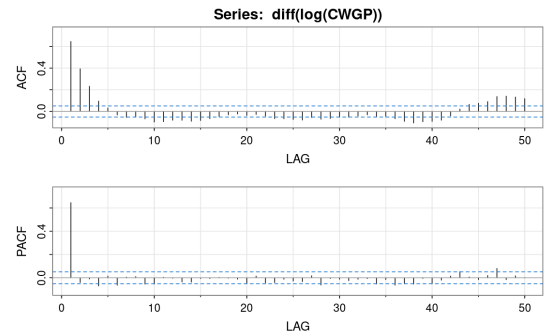


Figure 4: ACF and PACF of our transformed data

The data looks like something we can work with now, presenting heteroscedacity and being centered around a mean, we have now met our stationarity assumptions. We can now use the tools in R's `astsa` package to analyse this stationary time series, and draw conclusions on how we can use a SARIMA model to analyse this data.

We now look at two of our powerful time series R functions: ACF and PACF. Our ACF (autocorrelation function) measures and plots the average correlation between data points in our time series and previous values for different lag quantities. Our PACF (partial autocorrelation function) differs in that each partial correlation controls for any correlation between observations of shorter lag lengths. In other words, the PACF measures the autocorrelation of two observations with the influence of the observations between them removed.

In our ACF plot, we see the strongest correlation or signal at lag one, after which we see exponential decay, similarly, for our PACF plot we see a peak at lag one, after which all signal vanishes. The behaviour of these two functions are key indicators of the order of an AR(1) process, where the current value of the series is based on the immediately preceding value and a stochastic error term. The ACF decays exponentially because the influence of a past value on future values decreases exponentially as the time gap increases in an AR(1) process. The PACF vanishes because the correlation between the series and its lagged values beyond lag 1 are fully explained by their mutual correlation with the first lag. Therefore, the PACF shows a significant correlation at lag 1, which is the immediate past value directly influencing the current value, and then drops to zero

or becomes insignificant after lag 1. This all seems to indicate that we can best predict next week's gas price by looking at last week's. Before creating our model however, we note that there seems to be a rise in the ACF at lag 48, around a year in lag. Let's turn our attention again to our original time series data.

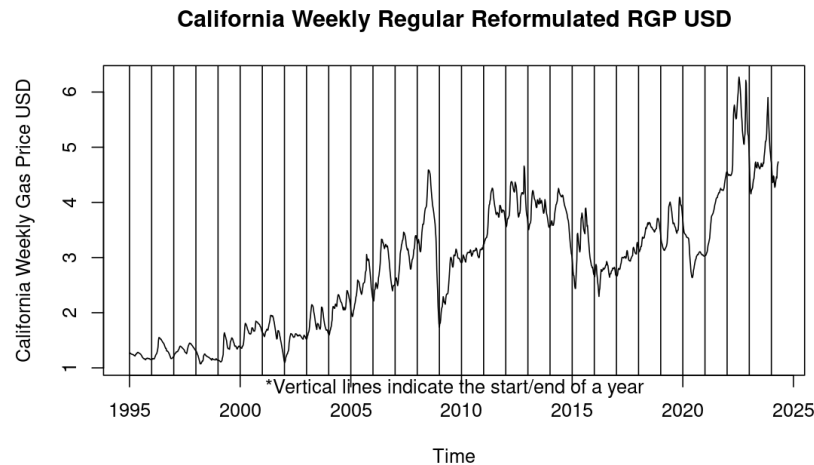


Figure 5: Original Time Series plot with divisions for different years

When observing our data split into years, to the naked eye, it seems that somewhat consistently we seem to observe peaks and troughs in similar locations – troughs towards the beginning and end of our different years, and peaks towards the middle. It would make sense to observe yearly seasonality in gas prices, as demand likely varies in similar ways over the year with different holidays and working periods. Changes in demand likely drive prices up and down consistently too. Let's look at our numbers once again and compare various models.

When ignoring potential yearly seasonality, our model as indicated by our PACF and ACF values is best described by $ARIMA(1,1,0)$: the first 1 denotes our AR(1) structure (x_t dependant on x_{t-1}); our second 1 denotes our original data differencing; and 0 doesn't acknowledge any MA structure. Our measure of AIC, Akaike Information Criterion, which is indicative of the level of fit and complexity of a given model, as suggested by R computes to -5.286752 . With approximately all (except the first 2) our Ljung-Box statistics computing to 0.

We proceed with different trial and error model testing, and note that we believe

$SARIMA(4, 1, 0) \times (1, 0, 1)_{52}$, most closely and concisely describes our data. $p = 4$, presents a AR(4) structure; $d = 1$, indicates our initial differencing; $q = 0$ no regular MA structure; $P = 1$ seasonal 52 week AR structure, and $Q = 1$ seasonal 52 week MA structure with $D = 1$, no seasonal differencing. We calculate the following coefficients:

AR1:	0.6519	p-value = 0
AR2:	-0.0344	p-value = 0.26
AR3:	0.0358	p-value = 0.24
AR4:	-0.0836	p-value = 0
SAR1:	0.9110	p-value = 0
SMA1:	-0.8542	p-value = 0
AIC = -5.304073		

These coefficient and p-values indicate that if we are trying to predict this week's gas prices, x_t , our x_{t-1} (last week); x_{t-4} (last month); x_{t-52} (last year) and ϵ_{t-52} (last year's error) are strong predictors. x_{t-2} , x_{t-3} 's high p-values seem to indicate that they aren't statistically significant, with our current tools however we can't make sense of our other values without them. The lower AIC value is indicative of a better fit than our previous model. It's hard to evaluate just how much better, but when contrasted with other AIC values computed for other potential models, the improvement seems to be significant. So we can summarise our $SARIMA(4, 1, 0) \times (1, 0, 1)_{52}$ as:

$$(1 + 0.652B - 0.034B^2 + 0.004B^3 - 0.084B^4)(1 - B)(1 + 0.911B^{52})x_t = (1 - 0.854B^{52})\epsilon_t$$

We turn to the residuals to further analyse the power of our model.

The ACF residuals, Figure 6, for the most part remain within our 95% confidence bounds, indicating that we have captured the majority of the signal provided by our time series. Our relatively large Ljung-Box statistic p-values are likewise confidence inducing as we'd expect for such a large sample size (over 1500 observations) to compute to 0 if our model weren't a good fit.

The variance of our model, built on logged data is σ_ϵ^2 , as estimated by R, is computed to be 0.0002876406. Considering that when we retrieve our forecast back we

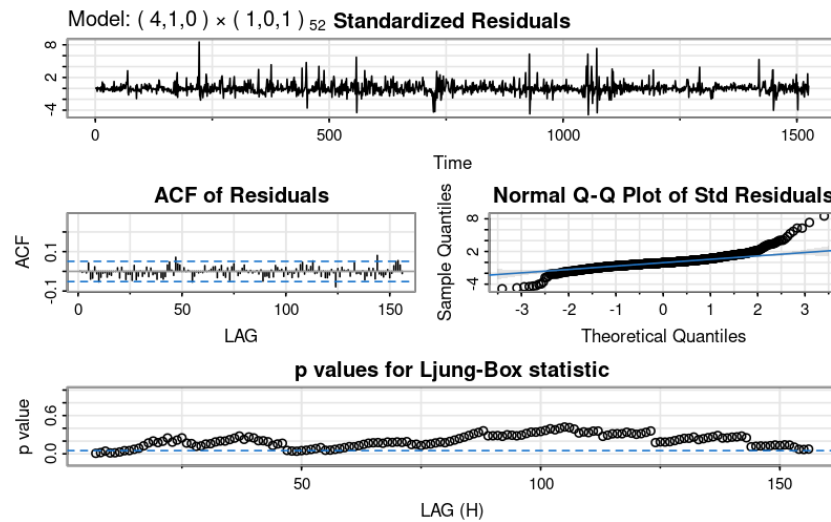


Figure 6: Residuals of our SARIMA Model

have to take the exponent of our computed values, this variance computes to being quite big. Let's look at what this variance looks like in our forecast:

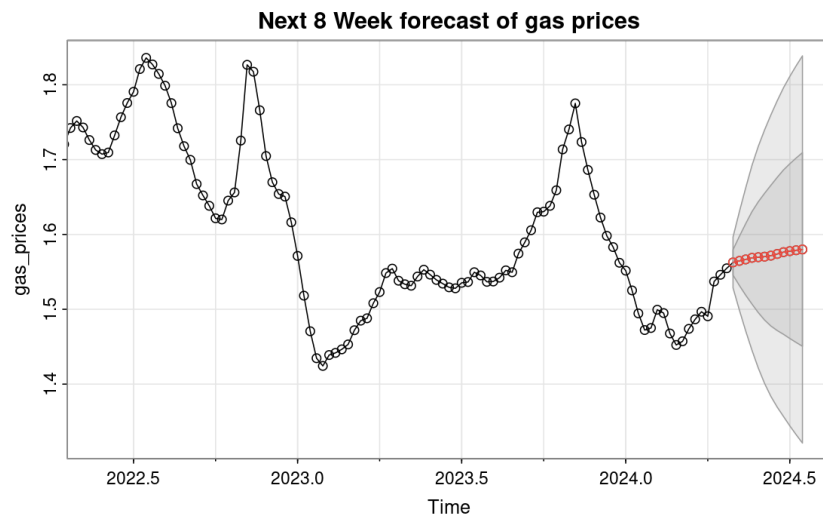


Figure 7: Caption

Even when only making a forecast for the next 8 data points, weeks, we can see that our variance and therefore margin of error explodes as we lose the most significant signal, that from the preceding week and belonging to our AR(1) at an exponential rate. Our one standard deviation error bounds (darker grey area) aren't terrible, however, 2 standard deviations away, lighter grey, our 95% bounds practically encompass more than the range of values we have seen for the last two years. The range of error indicates the high volatility of gas prices in California, showing us just how hard it is to make accurate

gas price predictions in the state.

For the 25th of March 2024, our model calculated 1.5626081 for the logged value of the retail gas price. Transforming this estimate gives us $e^{1.5626081} \approx \$ 4.771$, which is higher than the preceding week's value \$ 4.734 the actual value recorded by EIA was 4.801, we can evidently see an error in our prediction... but not bad at all! At least we predicted a rise in the retail gas price!

4 Conclusion

Using our time series tools we have created a $SARIMA(4, 1, 0) \times (1, 0, 1)_{52}$ model to forecast the retail gas prices of reformulated gas prices in California. Within this analysis, we discovered that the best indicators of this week's gas prices, are last week's and month's, as suggested by our AR(4) model. We noted that there seems to be a yearly AR and MA seasonal trend in our data, so there is also use in looking at the trend of prices a year ago when forecasting.

It seems relatively sensible that our closest observation, last week's, is the best indicator of this week. Similarly, it also makes sense that we'd see some predictive power four weeks ago; monthly pay cycles may dictate how much and when gas is purchased for families that live check to check. As approximately 20% of California's workforce live off the minimum wage, it makes sense that these pay cycles would have some sort of impact on the demand of gas and therefore on the price every month. When it comes to yearly trends, we may suspect that at certain times of the year, there are "abnormal" changes in gas-price demand, regular holiday seasons may indicate higher gas demand as people living in the sunny state may travel more. Summer holiday periods for example, may see a change in these prices. However, these values may not have as strong of a coefficient or impact on the model because other periods of the year may not share as strong of trends. Mentioning this opens a new Pandora's box, and leads us to the discussion of the weaknesses in this analysis, as it may indicate a lack of stationarity. Our previous statement seems to suggest that it matters where in the time series we are not only how

far apart any of our two observations are. Cleaning this issue up requires tools beyond the scope of this paper.

Even though we are content with our model, we have also shown that there is a high variability associated with our model, which is unsurprising given previous literature on the matter. This shows just how hard it is to predict gas prices. As George Box would say, “*All models are wrong, but some are useful*”. Even with a “good” model, we anticipate a wide range of values which is extremely telling of the volatility of gas prices. My suggestion: unless you’re the one selling the gas, don’t make any bets on next month’s price.

References

- [1] CA Gas Prices, Borenstein, 2023, <https://haas.berkeley.edu/energy-institute/research/borenstein-ca-gas-prices/>.
- [2] What Drives California's Gasoline Prices?, California Energy Commission, Sep 2022, <https://www.energy.ca.gov/data-reports/energy-insights/what-drives-californias-gasoline-prices>
- [3] California bumps up its minimum wage, Rose Dickey, 2023 , <https://www.axios.com/local/san-francisco/2023/01/06/california-minimum-wage-increase>
- [4] Methodology for EIA Weekly Retail Gasoline Price Estimates, EIA, 2024 , https://www.eia.gov/petroleum/gasdiesel/gas_proc-methods.php
- [5] What drives volatility in natural gas prices?, Abebe Hailemariam, Russell Smyth, Energy Economics, Volume 80, 2019, <https://www.sciencedirect.com/science/article/pii/S0140988319300623>